KOLHAPUR INSTITUTE OF TECHNOLOGY'S

COLLEGE OF ENGINEERING, KOLHAPUR

(EMPOWERED AUTONOMOUS)

Department of Computer Science & Engineering

MINI PROJECT SYNOPSIS

ON

MediMate : A smart companion for prescription and medication journey

Submitted By

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Under the guidance of

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Project Title:

MediMate: A smart companion for prescription and medication journey

Introduction:

In healthcare, many patients face difficulty in understanding handwritten prescriptions due to unclear handwriting and complex medical terms. This problem is more serious for elderly and rural users who may not be familiar with English or medical instructions. Existing digital prescription systems are available, but they are not user-friendly and lack support for regional languages and voice assistance.

Optical Character Recognition (OCR) technology is used to convert prescription images into text, but it often produces errors due to poor handwriting. Medicine information applications provide details about drugs, but they do not integrate prescription understanding, translation, and reminders into one system. SMS reminder systems exist but usually require manual setup.

The Smart Prescription Reader and Reminder System is a web-based application designed to solve these problems. It allows users to upload prescriptions, extract and process text using OCR and AI techniques, and convert it into a structured and easy-to-understand format. The system provides regional language support, voice output, medicine details, and automatic reminder scheduling, making it especially useful for elderly and rural users.

Objective :

- To develop a system for converting handwritten prescriptions into digital format
- To extract medicine name, dosage, duration, and frequency using OCR and AI
- To provide regional language support (Marathi/Hindi)
- To assist users through voice output using Text-to-Speech
- To provide detailed medicine information for safe usage
- To automate reminder scheduling for medicine intake

Feasibility Study :

Technical Feasibility:
The project is technically feasible using OCR tools like Tesseract, NLP techniques, and Machine Learning models such as Logistic Regression or Random Forest. Web technologies like React and Node.js can be used for development.

Operational Feasibility:
The system is easy to use with simple steps like login, upload, and viewing results. It is suitable for all users, including elderly and rural people.

Economic Feasibility:
The project is cost-effective as it uses open-source technologies and does not require additional hardware.

Need and Significance:
The system improves medication safety by making prescriptions understandable and accessible. It is especially helpful for users who have difficulty reading or understanding prescriptions.

Methodology and Technologies :

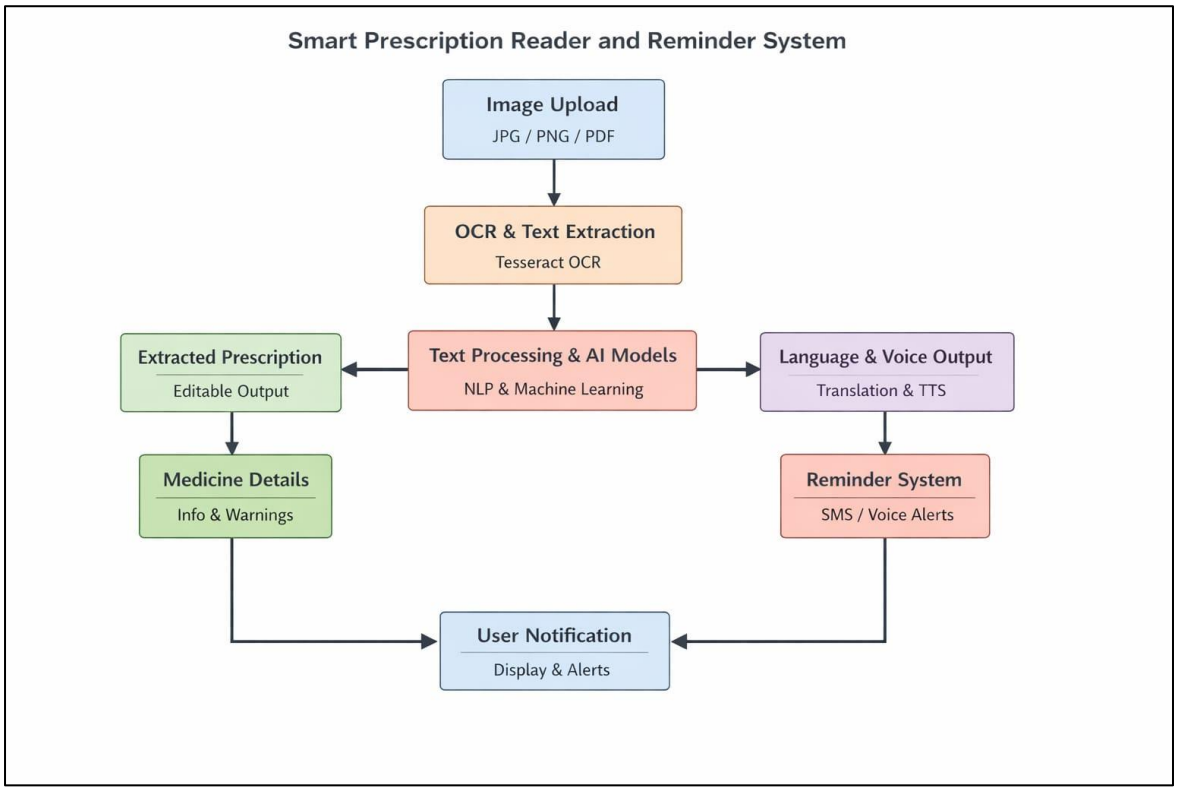
The system follows a structured pipeline. First, the user logs in using a name or phone number and password for secure access. The user then **uploads a prescription in JPG, PNG, or PDF** format. The system processes both scanned and photographed prescriptions.

OCR technology using **Tesseract and Python pytesseract** is applied to extract text from the uploaded document. The extracted text is cleaned and formatted using Python libraries such as **regular expressions and pandas** to remove noise and errors.

Natural Language Processing (NLP) techniques such as tokenization, named entity recognition, and text normalization are used to identify key components like **medicine name, dosage, duration, and frequency.**

An AI-based processing module extracts structured data and verifies it using a trained dataset and drug database. The system generates a clean and readable **English prescription output**.

The system also provides **regional language** conversion using translation APIs and converts text into speech using **Text-to-Speech technology**. Users can listen to the full prescription or individualmedicines.



Project is an Application Based Project

a) Software / Hardware Requirements

1. Software Requirements:

Frontend: HTML, CSS, Bootstrap, JavaScript

Backend: Node.js, Express.js

Database: PostgreSQL

OCR: Tesseract OCR, Python pytesseract

Machine Learning: Scikit-learn

NLP: Python (NLTK / spaCy)

SMS API: Twilio / Fast2SMS

Translation API: Google Translate API

Text-to-Speech: Web Speech API

Operating System: Windows

2.Hardware Requirements :

- RAM: Minimum 8 GB
- Storage: 20 GB free space
- Internet connection

b) Benefits of the Project for the Society

- Makes prescriptions easy to understand
- Reduces medication errors
- Helps elderly and rural users
- Provides multilingual and voice support
- Improves medicine adherence through reminders
- Enhances patient safety and awareness

References :

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